PARTIALLY AS CONSTRUCTED

R03 5

415V 3 PHASE + NEUTRAL 50Hz 2500A FORM 4 FAULT LEVEL 50kA FOR 1 sec.

CONTINUED DWG 34\E\0021-001

F02 1 — 50A — AFS — F35PP448-VS VSD — M 15 kW — 1RL3
F35-PP-448 — F1 CIRCUIT 2 — FEED PUMP — REF DWG 2000-F34-SCM-EL-10003 — TO SUBSTATION COMMS RACK

F02 2 — 20A — AFS — F35PP435-VS VSD — M 5.5 kW — 1RL2
F35-PP-435 — UFG CIRCUIT 1 — FEED PUMP — REF DWG 2000-F34-SCM-EL-10002 — TO SUBSTATION COMMS RACK

F02 3 — 100A 160A — AFS — F35-PP-61BVS VSD — M 30 kW
F35-PP-61B — ULTRA FINE GRINDING CONCENTRATE THICKENER U/F PUMP B — REF. DWG. 34\E\0041-018 — TO PROCESS CONTROL CUBICLE

F02 4 — 100A 160A — CA-60 — AFS — F35-PP-61AVS VSD — M 30 kW
F35-PP-61A — ULTRA FINE GRINDING CONCENTRATE THICKENER U/F PUMP A — REF. DWG. 34\E\0041-010 — TO PROCESS CONTROL CUBICLE

F02 5 — 315A 400A — CA6-210-EI @ 350A — AFS — F35-PP-63AVS VSD — RTD — M 110 kW — NOTE 2. — 2
F35-PP-63A — ULTRA FINE GRINDING CYCLONE FEED PUMP A — REF. DWG. 34\E\0041-010 — 2000-F34-SCM-EL-10005 — TO PROCESS CONTROL CUBICLE

F03 1 — AFS — SPARE

F03 2 — 63A — AFS — BYPASS SWITCH & TRANSFORMER — 240V — 110V
F34-XE-02 — SUBSTATION No.10 240V UN INTERRUPTABLE POWER SUPPLY — REF. DWG. 34\E\0041-004 — 11
F34-XE-02F — SUBSTATION No.10 240V/110V PROTECTION EQUIPMENT SUPPLY — REF. DWG 20\E\0421-002 & 20\E\0001

F03 3 — 80A 100A — AFS — SPARE

F03 4 — 125A 200A — CA-72 — AFS — F34-PP-01VS VSD — M 37 kW
F34-PP-01 — ULTRA FINE GRINDING MILL FEED PUMP — REF. DWG. 34\E\0041-010 — TO PROCESS CONTROL CUBICLE

F03 5 — 315A 400A — CA6-210-EI @ 350A — AFS — F35-PP-63BVS VSD — RTD — M 110 kW — NOTE 2. — 2
F35-PP-63B — ULTRA FINE GRINDING DESLIME CYCLONE FEED PUMP B FEED PUMP B — REF. DWG. 34\E\0041-010 — 2000-F34-SCM-EL-10005 — TO PROCESS CONTROL CUBICLE

F04 1 — AFS — SPARE
34-TR-03 — 110V SUPPLY MCC 34-SW-01 — REF DWG 34\E\0041-001 ONWARDS

F04 2 — 20A 32A — 415V/110V 2KVA — 16A — AFS
110kW DRIVE PANEL 35-AG-61B — 100/1A — Q102 — F102 — K113 — M — 2
35-AG-61B — FILTER FEED TANK No.2 AGITATOR — REF. DWG. 2000-F34-SCM-EL-10102

F02 2 — 63A — 50A

R02 4 — 32A 100A — AFS — F34CR02-IS — 20A — CRANE — 4 kW
F34-CR-02 — ULTRA FINE GRINDING DAVIT CRANE — REF. DWG 34\VE\xxx

R02 5 — 16A 100A — AFS — F34CR01-IS — 20A — CRANE — 2.2 kW
F34-CR-01 — ULTRA FINE GRINDING MONORAIL CRANE — REF. DWG 34\VE\xxx

R03 1 — 32A 63A — AFS
F34-RC-01 — ULTRA FINE GRINDING LIQUID RESISTANCE STARTER CONTROL PANEL — REF. DWG 34\E\0041-011 — REF. DWG 34\E\0021-001

R03 2 — 50A 63A — AFS — M 11 kW
F34-PS-02 — ULTRA FINE GRINDING CYCLONE AREA SUMP PUMP — REF. DWG 34\E\0041-005

R03 3 — 32A 63A — AFS — M 15 kW
F34-PS-01 — ULTRA FINE GRINDING MILL AREA SUMP PUMP — REF. DWG 34\E\0041-005

R03 4 — 16A 63A — AFS — M 1.5 kW
34-SC-01 — UFG VIBRATING SCREEN — REF. DWG 34\E\0041-xxx

R03 5 — 63A — AFS — F34PP443-VS VSD — M 22 kW — 1RL3
F34-PP-443 — UFG MILL 1 PRODUCT PUMP — REF DWG: 2000-F34-SCM-EL-10001 — TO SUBSTATION COMMS RACK

R03 6 — 50A — AFS — F35PP449-VS VSD — M 15 kW — 2
F35-PP-449 — UFG CIRCUIT 3 FEED PUMP — REF. DWG: 2000-F34-SCM-EL-10004 — TO SUBSTATION COMMS RACK

R03 7 — 10A 63A — AFS — M xx kW
F34-FV-01 — UFG VIBRATING FEEDER — REF. DWG 34\E\0041-xxx

R03 8 — 6A 63A — AFS — M 0.75 kW
F34-HE-01 — ULTRA FINE GRINDING MILL DRIVE HEATER — REF. DWG 34\E\0041-004

R04 1 — SPACE

R04 2 — 10A 63A — CA7-9 — AFS — M 1.5 kW
F34-FN-01 — ULTRA FINE GRINDING GEARBOX OIL COOLING FAN — REF. DWG 34\E\0041-004

R04 3 — 100A 100A — CA7-60 — 80/5 W — A/mA — AFS — M 30 kW — 2
F35-AG-62 — ULTRA FINE GRINDING PLANT FEED TANK AGITATOR — REF. DWG. 34\E\0041-006 — TO PROCESS CONTROL CUBICLE

R04 4 — 125A 60A — CA-76 — 125/5 W — A/mA — AFS — M 37 kW
F35-PW-62 — ULTRA FINE GRINDING PLANT FEED PUMP — REF. DWG 34\E\0041-006 — TO PROCESS CONTROL CUBICLE

R04 5 — 250A 400A — CA6-210-EI @ 350A — NOTE 3 — 250/5 W — A/mA — AFS — RTD — M 110 kW — 2
F35-AG-61B — ULTRA FINE GRINDING FILTER FEED TANK AGITATOR — REF. DWG. 34\E\0041-007 — TO PROCESS CONTROL CUBICLE

SOFT STARTER — 2 — 1RL1

| | |
|---|---|
| Approved | X |
| Approved as Noted | |
| Not Approved | |
| For Information Only | |

Signed: C Laird — Date: 07/08/2026

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

LEGEND

- COMBINED FUSE SWITCH
- CONTACTOR
- OVERLOAD
- VSD — VARIABLE SPEED DRIVE
- RTD
- THERMISTOR
- LOCAL CONTROL STATION
- M — MOTOR
- BUCHHOLZ PROTECTION
- PQM — POWER QUALITY METER
- LINK
- FUSE
- UPS
- OVERCURRENT RELAY WITH TIME LAG
- I INST — INSTANTANEOUS EARTH CURRENT RELAY
- A mA — AMP SIGNAL CONVERTOR
- 5 20A — GPO
- HEATER
- CONTROL TRANSFORMER
- TRANSFORMER
- WITHDRAWABLE BREAKER HEATER
- DISTRIBUTION BOARD

LEGEND

- STAGE 1 - NEW
- STAGE 1 - REMOVE
- STAGE 1 - DECOMMISSIONED (OTHERS)
- STAGE 2 - NEW
- STAGE 2 - REMOVE
- STAGE 2 - DECOMMISSIONED (OTHERS)

NOTES

1. ALL AFS (ARC FLASH SENSORS) ARE CONNECTED TO THE ARC FLASH RELAY VIA I/O UNITS.
2. **VSD OUTGOING FIELD CABLES DISCONNECTED AND MADE SAFE AT BOTH ENDS.**
3. **BUCKET R04/5 EQUIPMENT IS NOT TO BE MODIFIED. BYPASS THE CONTACTOR ONLY AND LEAVE INSITUE. NEW CABLE TERMINATES TO LOAD SIDE OF EXISTING FUSED SWITCH**

**KCGM GROWTH PROJECT**

DRAWING NO: 2000-F34-SLD-EL-10102

| | | |
|---|---|---|
| 1RL3 | 20/07/26 | AS CONSTRUCTED FOR SP1 |
| 1RL2 | 04/06/26 | AS CONSTRUCTED FOR F34.20.01 |

# 2000-F34-SLD-EL-10102_Rev2

Substation no 10 415V MCC 34-SW-02 - single line diagram